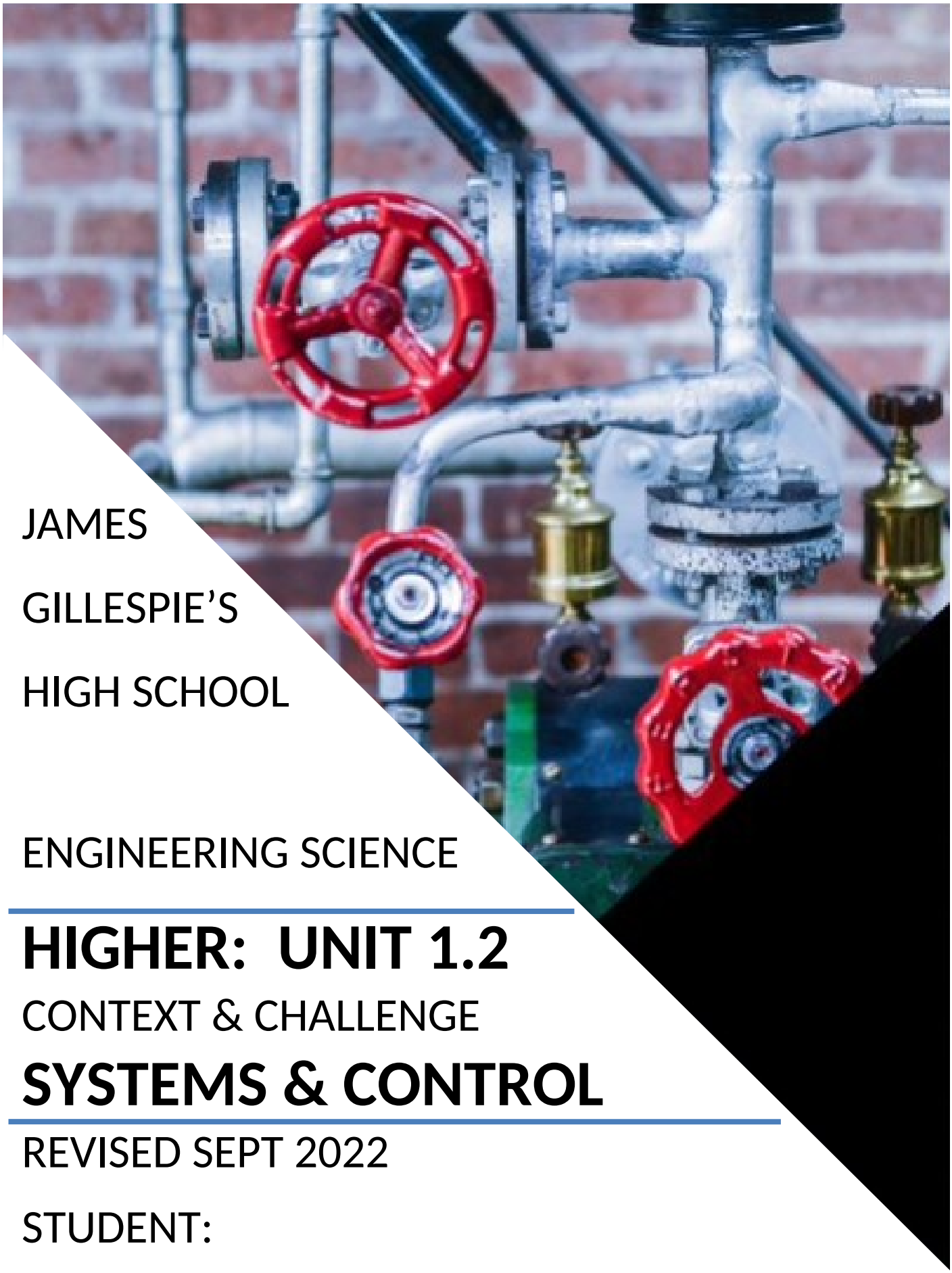


JAMES

GILLESPIE'S

HIGH SCHOOL

ENGINEERING SCIENCE

HIGHER: UNIT 1.2

CONTEXT & CHALLENGE

SYSTEMS & CONTROL

REVISED SEPT 2022

STUDENT:

YEAR: TEACHER:

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Context and Challenge: Systems and Control

1. Course content:

Outcomes and Assessment Standards

There are main 3 outcomes in this unit and it has been divided into three booklets. By the end of this second booklet, you will have completed outcomes 1.1, 1.2 and 1.3. You will investigate engineering systems, problems and solutions, involving some existing and emerging technologies and consider implications relating to the environment, sustainable developments, and to economic and social issues.

The Context and Challenge unit Outcomes are:

1. Investigate engineered objects.
2. Investigate engineered challenges and relate these too key engineered concepts.
3. Describe some aspects of the impact of engineering.

Outcome 1

The candidate will:

1 Investigate engineered objects by:

1.1 Describing, using the systems approach, how some engineered objects work.
1.2 Identifying sub-systems and describing the function of each and how they interact.
1.3 Producing system diagrams to show sub-systems.
1.4 Carrying out energy audits.

Outcome 2

The candidate will:

2 Investigate engineering challenges and relate them to key engineering concepts by:

2.1 Identifying and describing how several different branches of engineering contribute to solving an engineering challenge.
2.2 Describing examples of the varied roles of engineers in designing, implementing, testing and controlling complex systems.
2.3 Modelling some aspect (related to one branch of engineering) of a solution to an engineering challenge.
2.4 Explaining how emerging technologies may provide improved solutions to engineering challenges.

Outcome 3

The candidate will:

3 Describe some aspects of the impact of engineering by:

3.1 Describing examples of social and economic impacts of engineering.
3.2 Describing some examples of environmental impacts of engineering.
3.3 Describing some ways in which engineering solutions contribute to tackling climate change.

Learning Intentions

- To learn about the varied role of Engineers in designing, implementing, testing and controlling complex systems.
- To learn the effects that engineering has on the world.
- To know about control diagrams for simple control systems.
- To learn about what the systems approach is.
- To learn about subsystems.
- To learn about energy forms.
- To learn about basic energy transformations.
- To know about open loop & closed loop control.
- To know what a feedback loop is.
- To learn about using error detector in a closed-loop system.
- The role of negative feedback in a system.
- To learn about closed-loop, two-state and proportional feedback.

Success Criteria

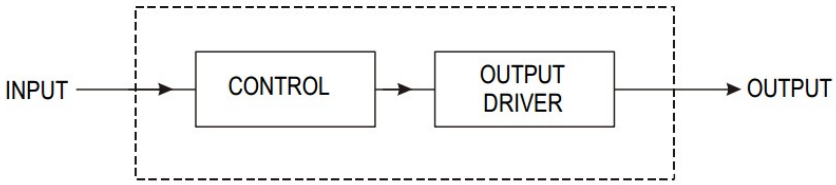
Use traffic lighting to record your progress. This will also help with revision as you will be able to see what areas you need to work on. Your teacher will sign off this page after speaking to you about your self-evaluation:

Red: I need to work on this topic further to understand it.
Amber: I need a bit of help to fully understand this topic.
Green: I am confident in my understanding of this topic.

- ☐ I can explain how emerging technologies can provide improves solutions to engineering challenges.
- ☐ I can identify basic energy transformations.
- ☐ I can draw and understand a system diagram, showing inputs and outputs.
- ☐ I can describe, using the systems approach, how some simple engineered objects work.
- ☐ I can identify the main sub-systems of an object.
- ☐ I can draw and understand sub-system diagrams.
- ☐ I can describe the operation of systems from reading their system/sub-system diagram.
- ☐ I can understand and create diagrams for simple control systems.
- ☐ I can understand the difference between automatic and manual control.
- ☐ I can understand how an error detector is used within control systems.

System diagrams

Most industrial product design is solved by the systems approach. This approach involves studying the desired function of the product, and then breaking this function down into a series of subsystems. When applied to control systems, a **Systems Diagram** is a useful way of visually representing the desired function of the system. The systems diagram is a form of block diagram than contains all the subsystems within a dashed box, called the systems boundary. The **systems boundary** indicates the extent of the control system. Anything outwith the systems boundary is considered the 'real world'.

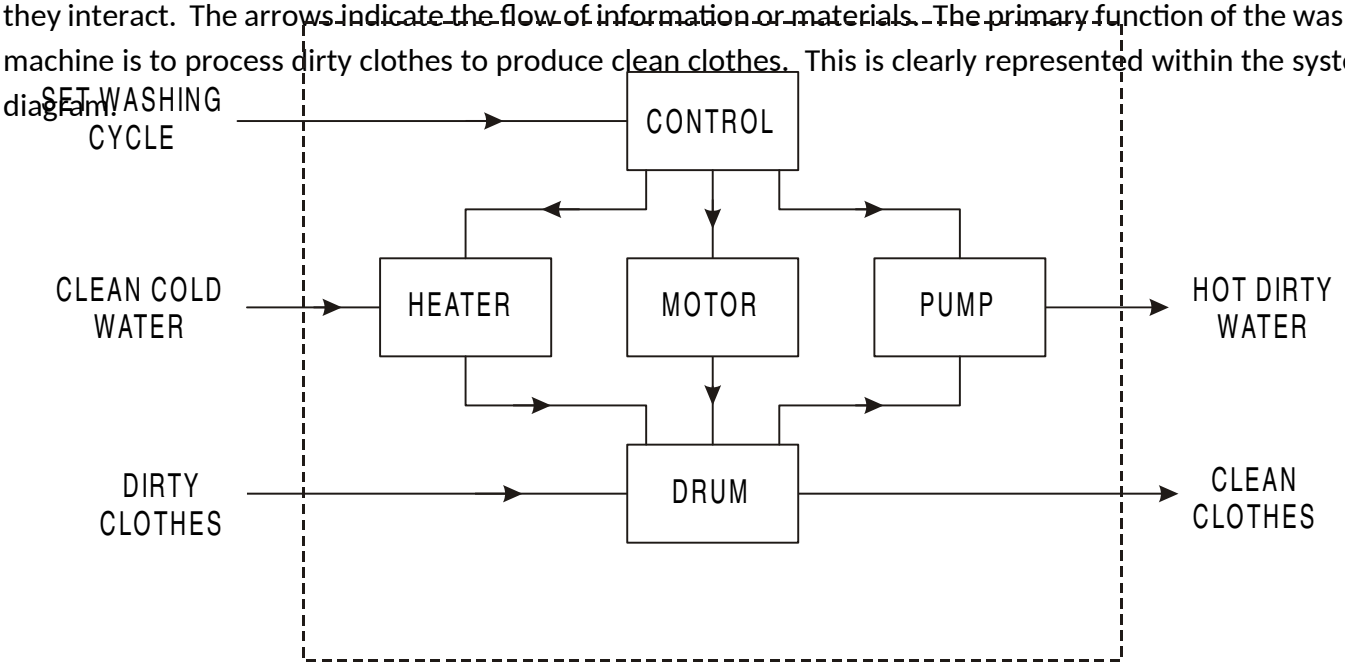


The "real world" input and output conditions of the system are shown as arrows entering, and leaving, the systems diagram. A product which has a fairly complicated control system is an automated washing machine. A simplified **universal systems diagram** of a washing machine is shown below. It is just simply, input, process and output.





This can be broken down to the **sub-systems diagram** shown below. It identifies the individual parts and how they interact. The arrows indicate the flow of information or materials. The primary function of the washing machine is to process dirty clothes to produce clean clothes. This is clearly represented within the systems diagram.



Open loop control



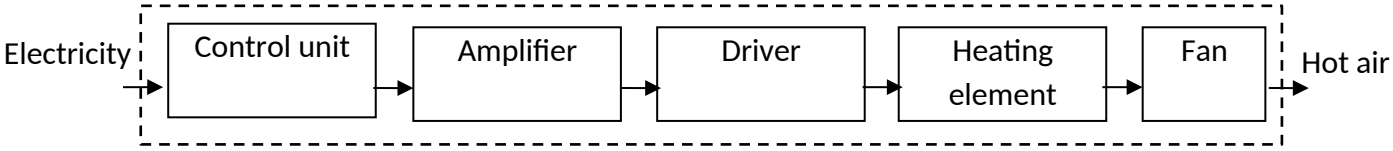
At the simplest level, a control system can process an input condition to produce a specified output. A good example of this type of open control loop system is a hand-held electric hairdryer. The heating element and fan motor are switched on when the appropriate switches are held down.

The air being blown out of the hairdryer is not temperature monitored or adjusted - the air is simply just blown out at whatever temperature the heater is capable of achieving.



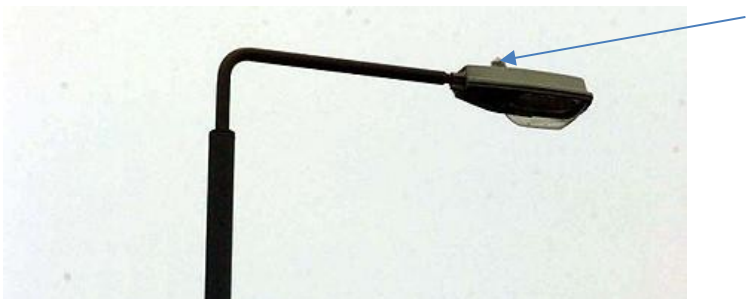
The problem with the 'universal system' diagram above is that it only shows the basic of operations of what goes in and out of the system and not much information can be learnt from what is actually going on. Only by breaking the system diagram down further do we get a true understanding of the internal workings.

By considering a sub system approach, the operation can be explained much easier as shown below.



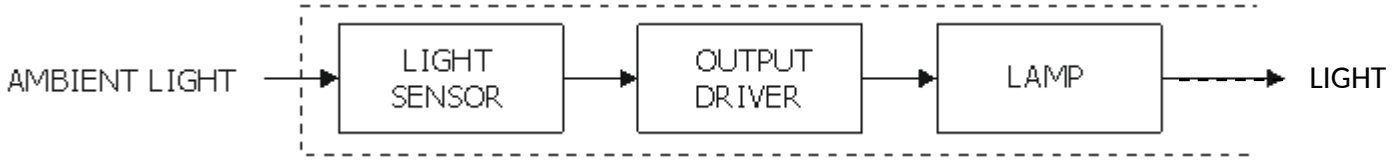
Once the button is pressed and electricity enters the system, it sends a signal into the control unit to trigger the start of the process. The signal that sends is now amplified by some form of amplifier, which in turn goes into a driver to increase the power to power the output transducer, in this case a heating element and fan which then produces the hot air.

Another example of an open loop system is an automatic street lamp.

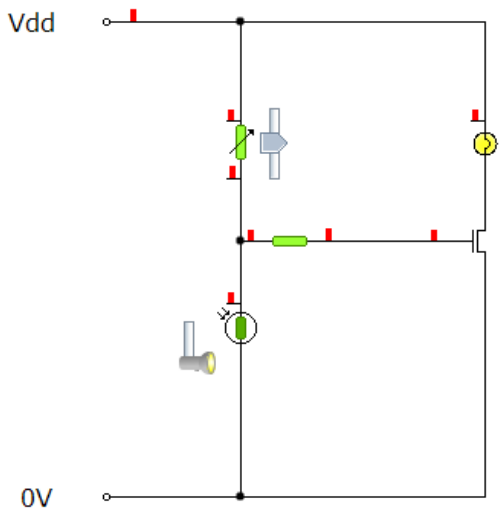




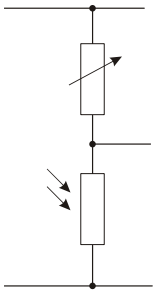
The systems diagram for the street lamp is shown below.



A simulation model of the street lamp can be represented by the following Yenka circuit diagram below:



How does it work?



1. The input to the circuit is light through the LDR. It is processed by the control subsystem to produce an electrical signal. In this example, the potential divider arrangement provides a varying voltage signal.

2. The electrical signal is processed by the output driver sub-system to produce the output. In this example, the output is light from the signal lamp

Open loop advantages:

An open loop control system represents the simplest and cheapest form of control. However, although open loop control has many applications, the biggest and basic weakness in this type of control lies in the lack of capability to monitor and adjust to suit the changing output requirements.

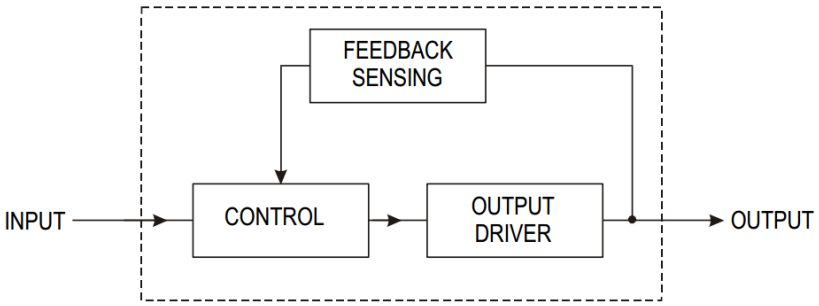
1. Complete system diagrams for the following:

A. Mobile phone.
B. Computer.
C. Hydro – electric power station.

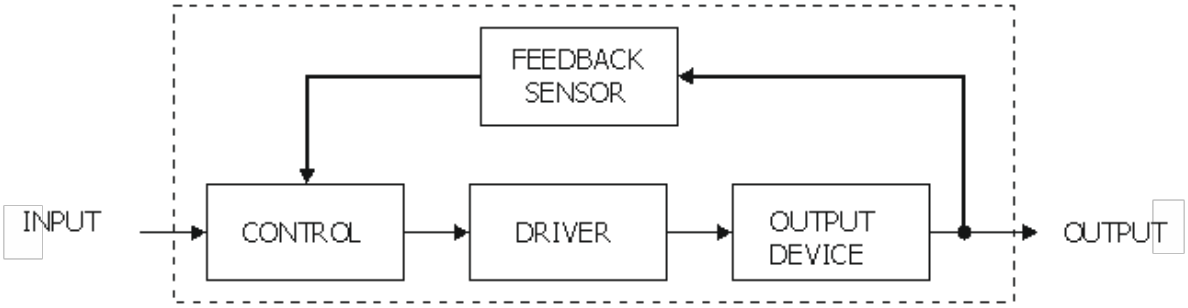
Closed loop control

The purpose of closed-loop control is to ensure that the output is maintained at, or as closely as possible to, the desired level.

All closed loop control systems include a **feedback sensing** subsystem within the systems diagram. The control subsystem will process the feedback signal by making a 'decision' on whether the state of the output should change.



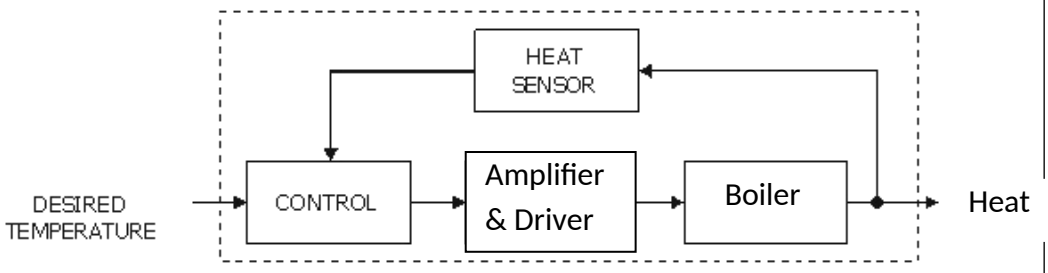
To operate the 'output device' correctly and efficiently, it requires a system 'driver' to increase any signal / voltage as shown below.



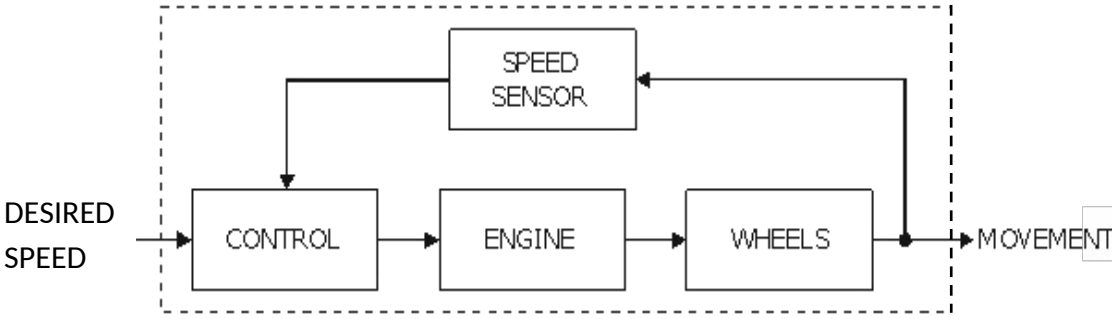
The central heating in most modern houses is controlled by a thermostat. It senses changes in temperature and turns off the heating system automatically when the required temperature is reached. Like wise, when the temperature level drops below the minimum acceptable level the heating system is automatically switched back on.

All closed loop control systems include a **feedback sensing** subsystem within the systems diagram. The control subsystem will process the feedback signal by making a 'decision' on whether the state of the output should change. To operate the 'output device' correctly and efficiently, the signal must be 'amplified' and then through a system 'driver' to increase any signal / voltage as shown below.



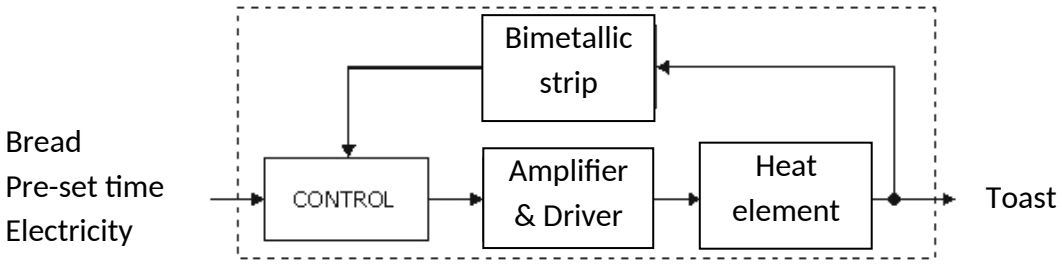


Another example of a closed loop control system is the cruise control on a car. This allows a car to travel at a constant speed, even when going up and down hills. In this case the feedback sensing system is a speed sensor (or tacho-generator) connected to a wheel. This gives out a higher voltage the faster it turns.



When the car goes up a hill it will start to slow down. The speed sensor feeds this information back to the control unit. The control unit then sends an appropriate signal to the engine to boost the power which will spin the wheels faster, bringing the speed of the car back towards the desired level.

Another example of a simple closed loop control system is an electric toaster. In this case the feedback sensing system is a mechanical bimetallic strip. This strip bends when heated - the higher the temperature the more the strip bends.



When the toaster cage is lowered into the toaster it compresses a spring, and the cage is 'hooked' into position by a catch attached to a bimetallic strip. A switch also connects the heating element, which starts to heat up. The heating element starts to cook the bread, and also starts to heat the bimetallic strip. When the bimetallic strip has been heated to the required temperature it bends enough to mechanically release the spring holding the cage down. This automatically ejects the toast and switches the heating element off.

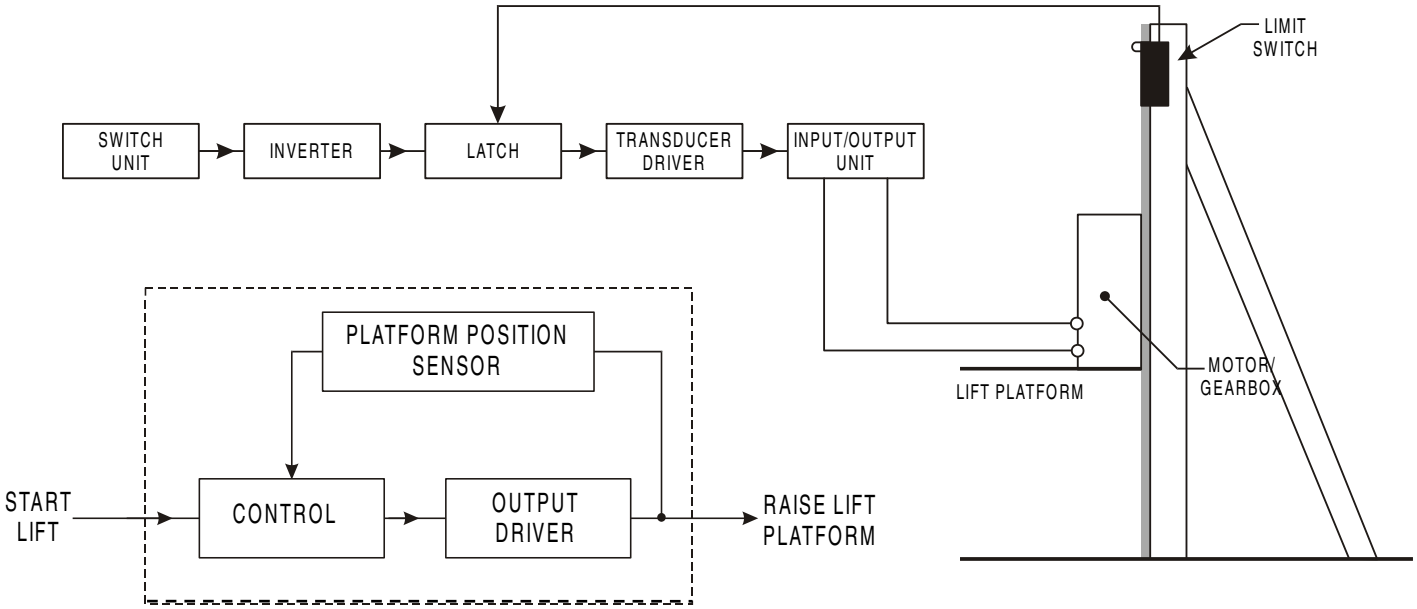
2. Explain what is meant by the term 'systems diagram'.

3. Describe the differences between an open loop and closed loop control system.

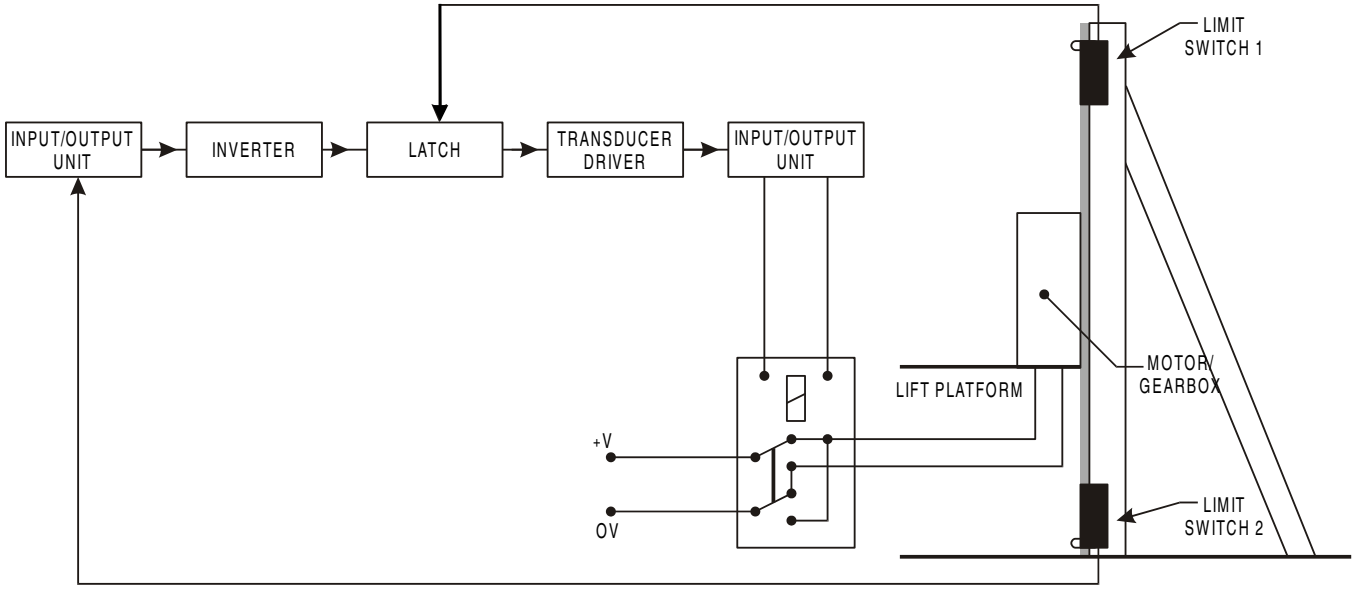
4. Describe the purpose of the feedback sensing subsystem in a closed loop control system.

5. The systems diagram for manual operated lift control is shown below. A limit switch was then added to make the system automatic. It was installed at the top of the lift so it detects when the lift platform has reached the top of the run.

a) Draw the revised automatic system diagram of the control system.
b) Draw a sub-system diagram of the control system.
c) Explain how the circuit would operate.



6. A second limit switch was added at the bottom of the lift. The output condition using this type of control does not have to be limited to on or off but can be applied to any two desired output states.



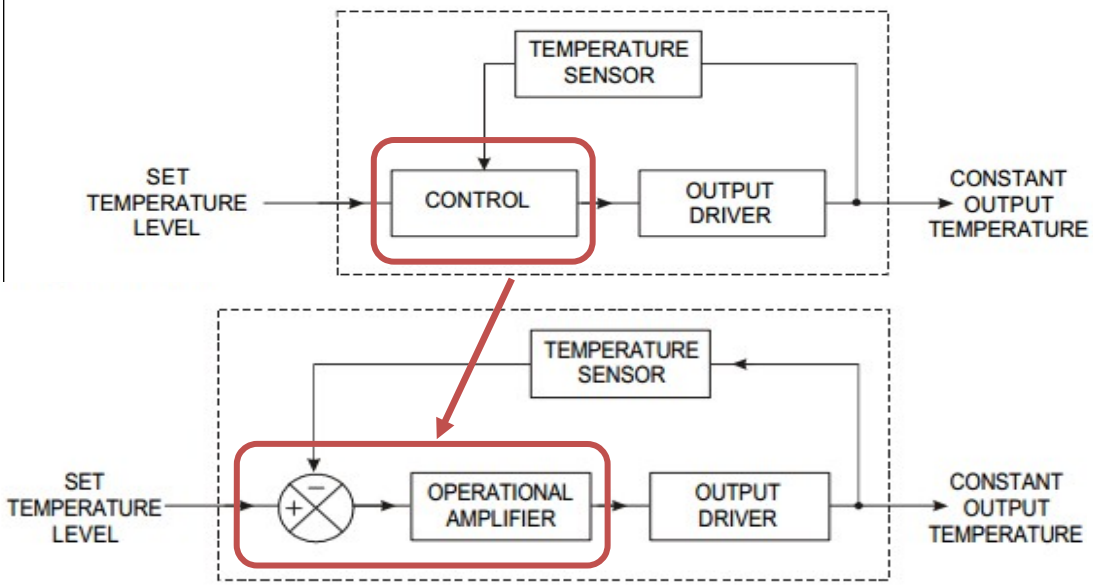
For the diagram shown

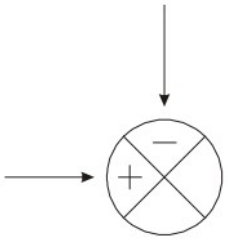
a) Draw a system diagram of the control system.
b) Draw a sub-system diagram of the control system.
c) Explain how the circuit would operate.

Control diagrams

The heating control system described on the previous page is represented using a Systems Diagram. However, in analogue control systems which involve processing analogue signals (heat, light and movement), op-amps are used. *You will learn more about how op-amps work in the Analogue electronics topic.*

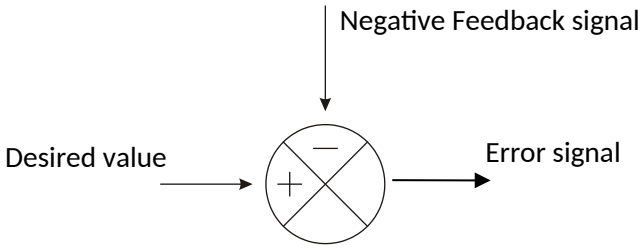
When op-amps are used it is usual to draw a control diagram.





The error-detection symbol is also used to indicate that the control involves comparing two signals.

Unlike National 5 Engineering Science education studies, at Higher, you must replace the 'control box' with the error detector. The error detector is a circle divided into 4 parts. Within two of the quadrants, there will be a 'positive' symbol (always at the 9 o'clock position) which represents the pre-programmed (set or desired input) and a 'negative' symbol which represents the negative feedback signal that is being monitored. For example, a room temperature set at 18°C. The other two quadrants must be left blank.

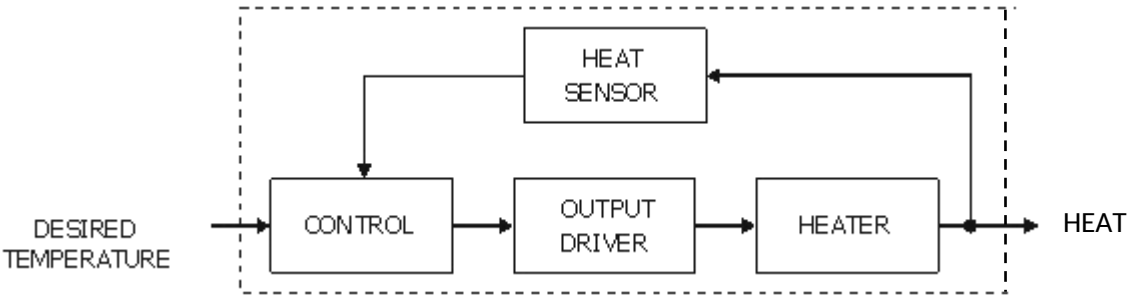


Basically the output value is being constantly monitored as the system operates via the feedback loop. The error-detection symbol is also used to indicate that the control involves two signals. If a difference exists between the actual value (output value) and the reference value (set input value), the system realises an error has occurred and will change the input into the system to reduce this output error to zero. Therefore closed-loop control systems are capable of making decisions and adjusting performance.

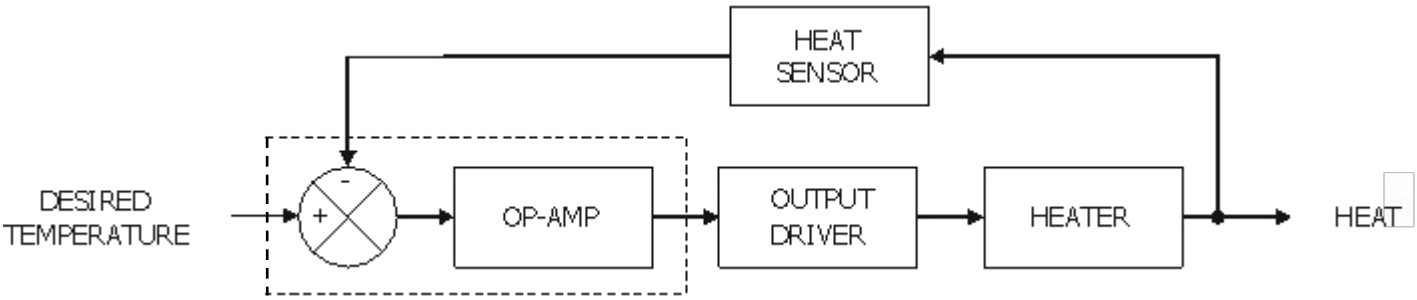
All closed-loop control systems include an error-detection function. In manual closed-loop control systems the brain is the error detector. In automatic closed-loop control systems the error detection is usually performed electronically by a device called an operational amplifier.

Control diagrams

The heating control system described previously can be represented by the following **systems diagram**.



However, when an op-amp is used in a control system, it is usual to draw a **control diagram** instead.



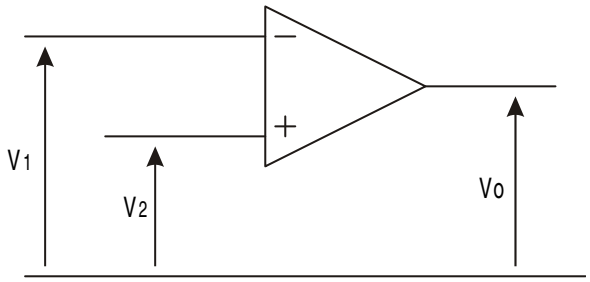
The control diagram breaks the generalised 'control' subsystem of the systems diagram into more specific blocks.

Analogue control systems

Two state control.

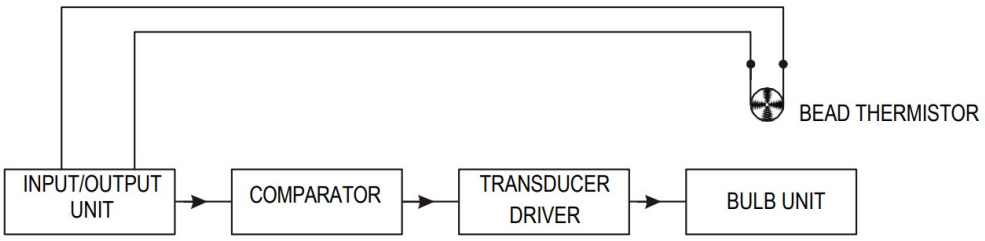
Many control systems involve processing analogue signals such as heat, light and movement. Therefore analogue closed loop control systems require analogue processing devices such as the **operational amplifier** (op-amp) shown below.

One of the most common control applications involves using the op-amp as a ‘comparator’. In it's simplest form a ‘comparator’ just compares two voltage signals, V1 and V2. If V1 is higher than V2, the output (Vo), is 'low', if V1 is lower than V2 the output is 'high'. A comparator provides ‘on/off’ or ‘two state’ control. The control diagram breaks the generalised 'control' subsystem of the systems diagram into more specific blocks.

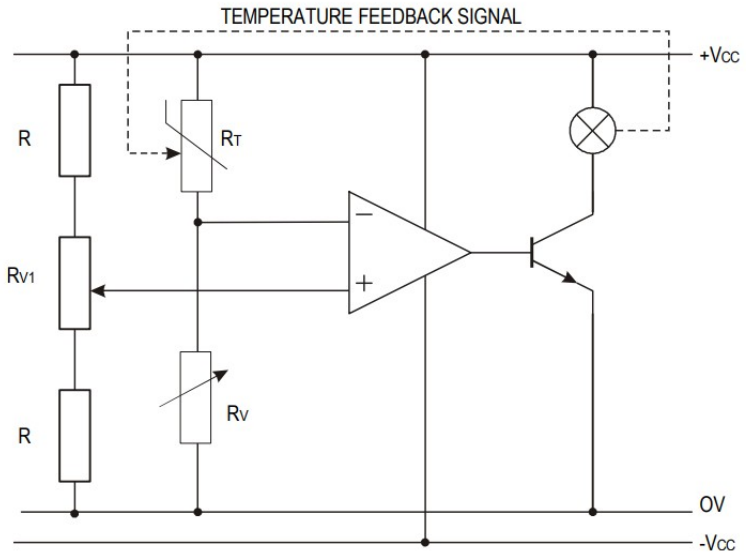


If Input of V1 higher than V2 = Output V0 is low
If input V1 is lower than V2 = Output V0 is high

7. The block diagram below simulates an automatic heating system. When the bulb heats up the temperature rise is detected by the bead thermistor, which sends a feedback signal back to the comparator.



A circuit diagram for the system is shown:



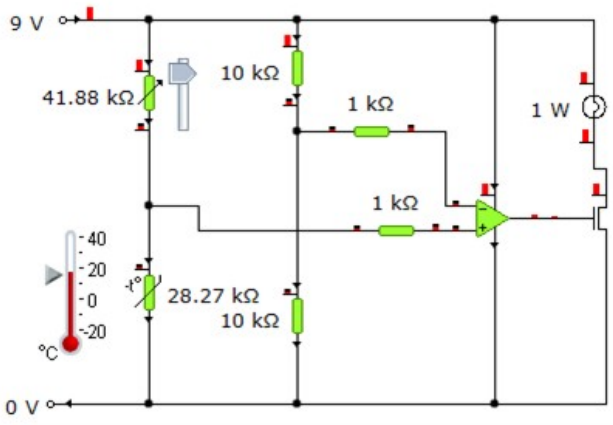
The variable resistor, RV1 (connecting to the non-inverting input of the op-amp) is used to set the reference level (or threshold). This sets the desired temperature of the bulb.

Build and test the circuit shown below, using Yenka. Calibrate the system to the following performance criteria:

- When the bulb heats above the reference level the thermistor should sense the temperature and send a signal to the comparator which will switch the bulb off.
- When the bulb cools below the reference level the bulb should switch on again.
- The system should operate continuously.

a) Describe how the circuit operates.

b) Explain clearly how you calibrated the system.

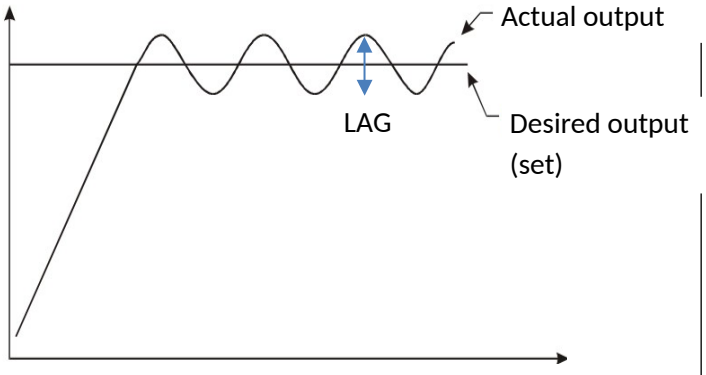


Types of feedback in a system

Negative feedback

The purpose of closed loop proportional control system is to ensure that the output is maintained, as closely as possible, to the desired output level.

The main problem with 'comparator' based control systems is that there is a time 'lag' built into their operation. This means that the desired output state is never actually reached. This means that the system is constantly trying to pull (reduce) the error back towards the desired output but it takes time to adjust. This type of control uses **negative feedback** to reduce the error.



Operation of a central heating system:

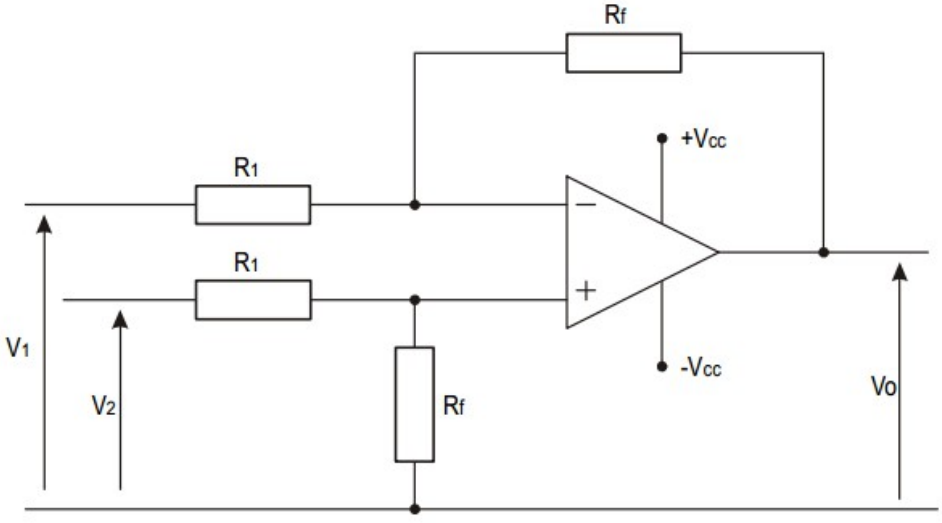
When the reference level ("set temperature") is reached the heating system will continue to heat the room for a short period until the heating element cools down. This raises the temperature in the room above the set or desired temperature level. When the temperature in the room cools down to the reference level the heating system will be switched on. Whilst the heating element is reaching its operating temperature the room continues to cool below the reference level.

Whilst in heating systems the presence of 'lag' does not present a problem, and in fact presents some advantages, in many other applications the inaccuracy of this form of control is not acceptable.

In the graph above, the heating element is still giving out heat well after it has reached the set temperature, even though it will have been turned off. In *proportional closed loop control*, the output from the system will depend on the **difference** between the set value and the actual value.

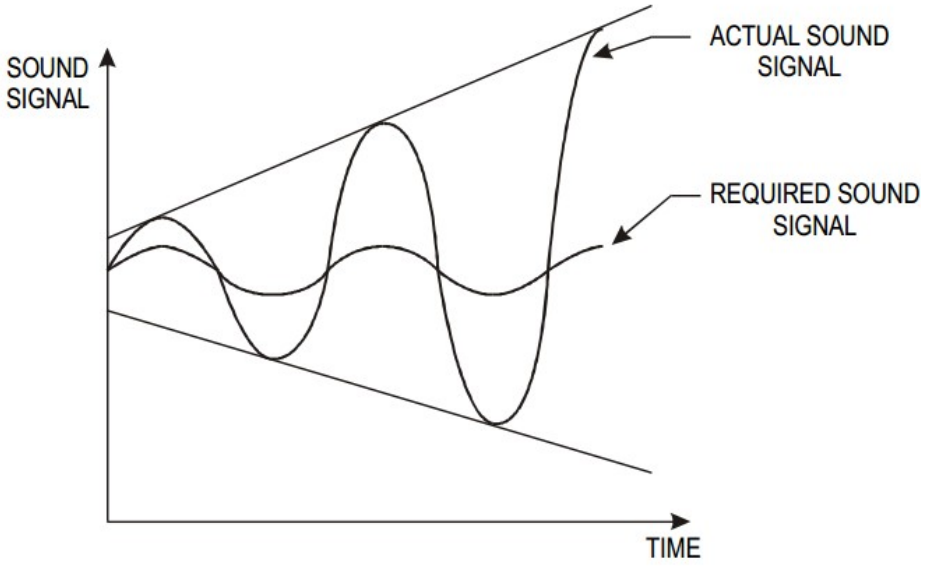
In the example above, it would mean that less power would be given to the heating element as the room approached the set temperature (and more power would be required if the room temperature decreased).

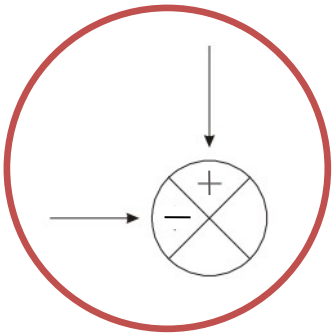
Proportional closed loop control systems are applied as a means of proving a more accurate form of analogue closed loop control. In order to achieve proportional closed loop control, we must use a '**Difference Amplifier**'.

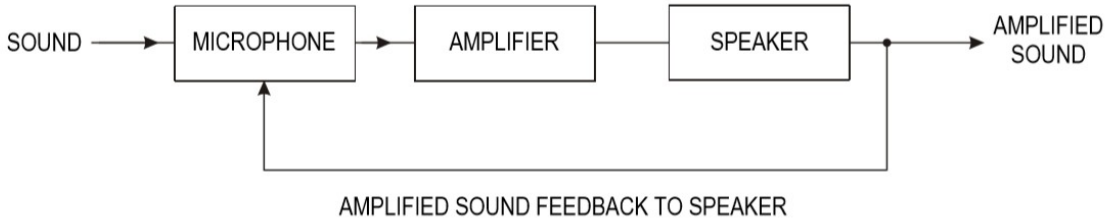


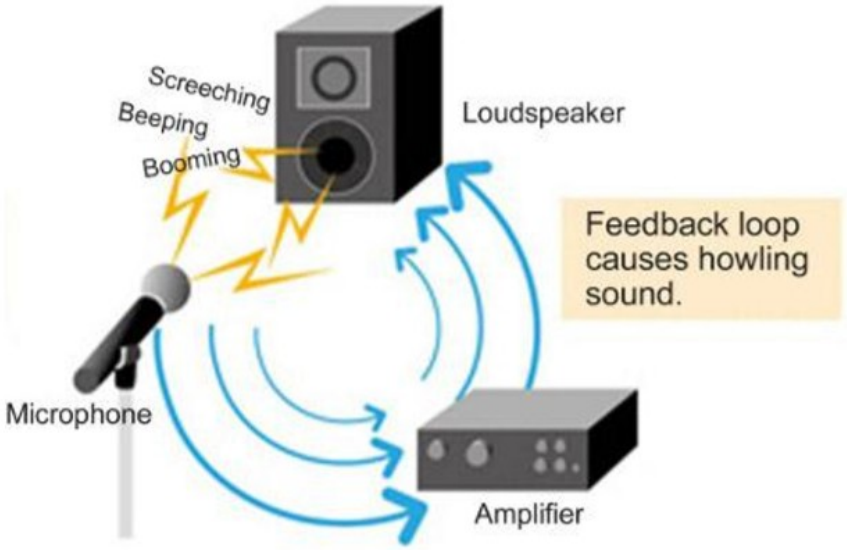
Positive feedback

The opposite effect can be created by reinforcing or increasing the error, as can sometimes happen with public address systems when the microphone is held too close to the speakers. A sound is picked up by the microphone, amplified, and then output through the speaker. The amplified sound is then picked up, re-amplified and so on. The net result is a high pitch sound, which can be represented by the graph below.







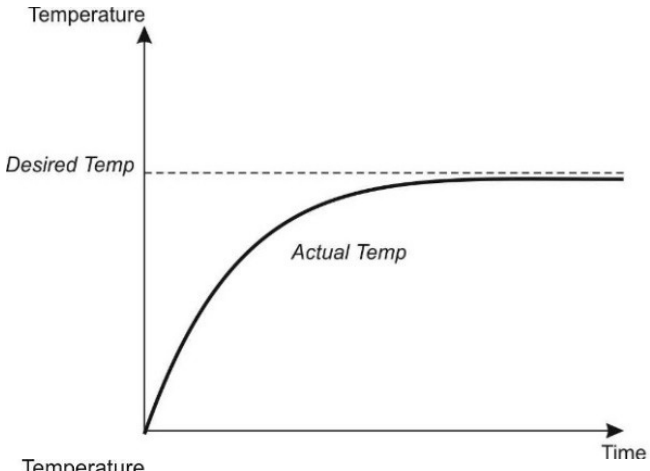


This is an example of **positive feedback**. Although positive feedback does have some useful applications, negative feedback is far more widely used in control systems.

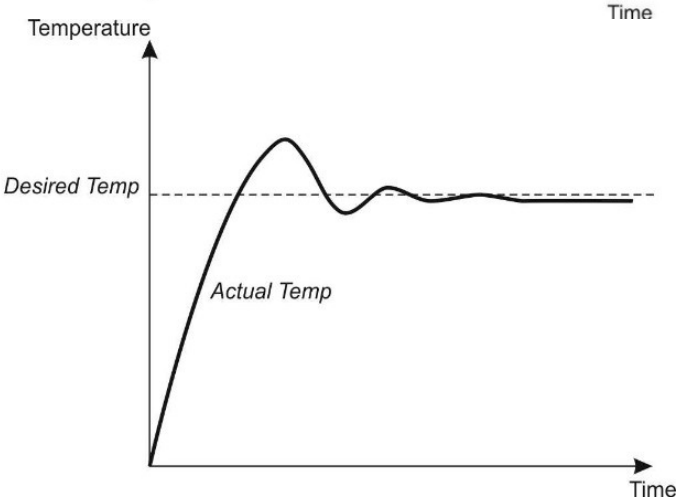
Proportional closed loop control

The 'difference amplifier' amplifies the difference between the reference level and the feedback signal. Therefore if there is a large error there will be a larger output signal than when there is a small error. This help prevents the 'overshoot' and 'time lags' seen in the 'comparator' based systems.

The first graph illustrates the effect when the gain of the op-amp is set **low**. Here, the actual temperature will rise quite slowly as the output of the system will not be so powerful. As it approaches the desired temperature, the difference between the actual and desired temperatures is less so the output gradually reduces. As a result, the rate of change of the actual temperature slows.



The second graph illustrates what happens when the gain of the op-amp is **high**. Here the system responds very quickly and the actual temperature initially overshoots the desired temperature. Some lag is observed but this gradually reduces until the temperature finally settles.



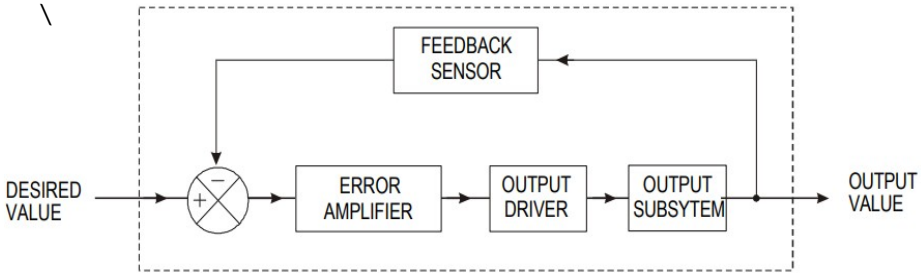
In each case the actual temperature settles fractionally below the desired temperature.

When working with proportional control, the aim is to adjust the gain so that we get a quick response but minimise the overshoot and time it takes to settle on the final desired output.

The circuit diagram below simulates the automatic water heating system within a washing machine.

Now our generic systems diagrams are starting to reflect the use of the op amp in the control area of the diagram:

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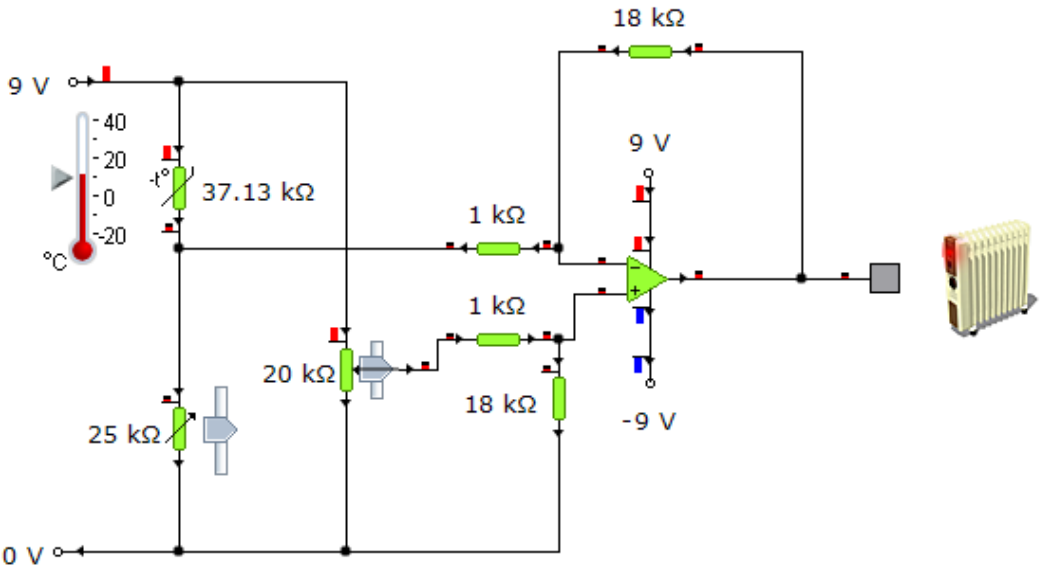


Sometimes, an operational amplifier symbol is shown instead of the block.

8.

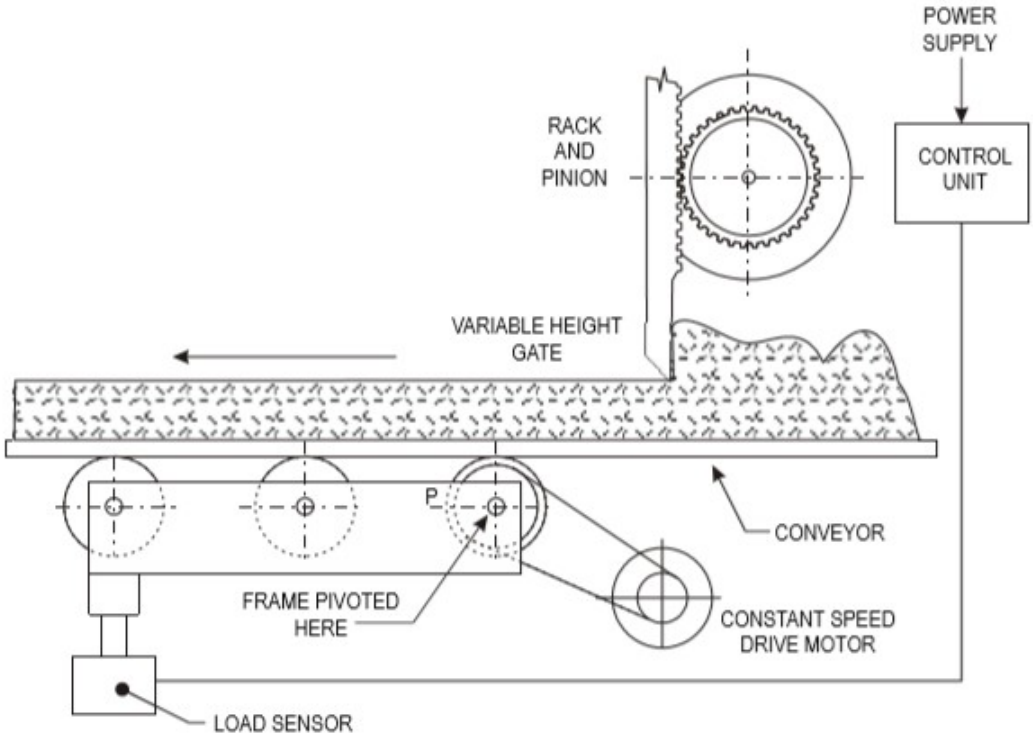
Build the circuit shown on Yenka. Observe the response of the system when the reference level (desired temperature) is altered (by adjusting the variable resistor).

a) Explain how the circuit above operates.
b) Draw a control system diagram for this system.
c) Explain in detail how the control system operates.



9.

A proportional control system is used to regulate the flow rate of coal onto a conveyer belt. The system should sense the weight of coal on the conveyer belt and automatically adjust the gate height to ensure that a constant flow of coal is supplied.



a) Draw a control systems diagram for this system
b) Explain the term 'proportional control'.
c) State the name of the configuration of op-amp used in a proportional control system.
d) Build a circuit diagram of the system on YENKA use a potentiometer as the load sensor
e) Sketch a suitable output driver circuit which could be used for the motor and give three reasons it is suitable for this application.

10.

A prototype for a new golf trolley requires a proportional control system that will keep it moving at a constant speed set by the player, even when moving up or down slopes. The speed will be monitored by a tacho-generator connected to one of the wheels.



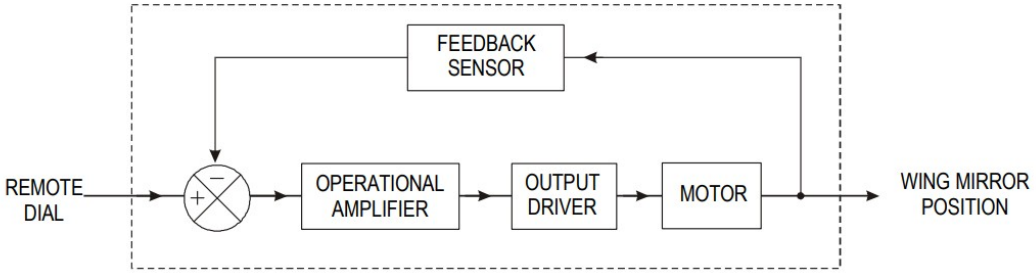
a) Draw a control diagram of the system.
b) Name the configuration of op-amp used in proportional control systems.
c) Draw a circuit diagram of the op-amp. (include appropriate input and output signals)

11.

A control system is required to move a car wing mirror. The driver will turn a dial indicating the desired position and the system will move the mirror to match it.



A control diagram of the system for rotational movement in the X-axis is shown below. Similar systems are used for the Y-axis.



a) With reference to the control system diagram, explain clearly how the system operates.
b) Name the type of control used in this system.
c) Name the configuration of op-amp required.
d) State two reasons why the op-amp cannot be used to drive the motor directly.
e) Name a suitable output driver which could be used with this system.
f) Draw a circuit diagram of the output driver.

12.

A street lamp is fitted with a light sensor that senses the surrounding natural (ambient) light intensity, but is not affected by the light from the street lamp itself. When the ambient light intensity is below a set level, the street lamp comes on fully. When the ambient light intensity is above a set level the street lamp switches off fully.

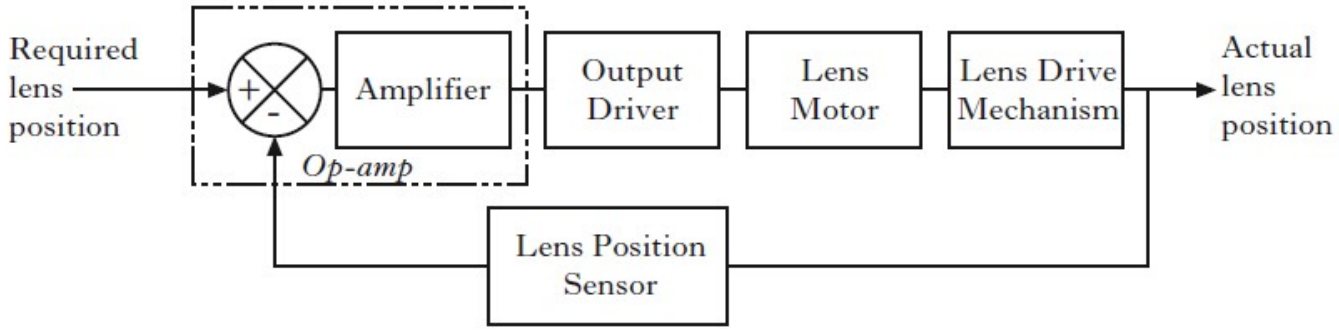
a) Draw a control diagram for this circuit.
b) State the full name of the type of control this uses.

13.

A quality-control system uses a portable conveyor belt for transporting items to be tested. The conveyor belt has a speed-control system allowing 3 different speeds to be selected. A tachogenerator fitted to the motor shaft sends a speed signal to the control system, and this is compared with a signal for the selected speed. The difference between the signals is amplified and sent to a driver which controls the motor.

a) Draw a control diagram for this system.
b) State the full name of the type of control this uses.

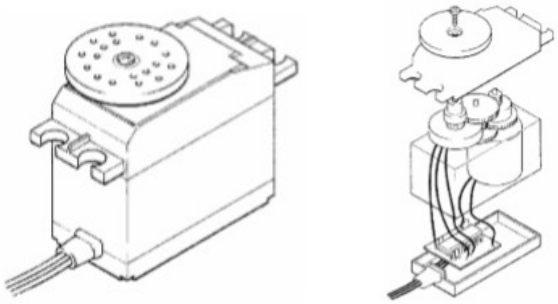
14. In order to focus an automatic camera, a control system must determine the required position of the camera lens using information from input sensors, and then control the small motors used to adjust the lens position.



a) Describe clearly how the system operates.
b) State which op-amp configuration would be most suitable to use in this system and justify your answer.

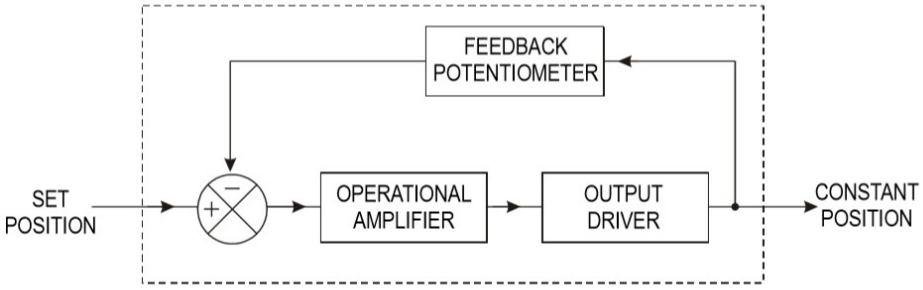
Positional control

A good example of positional control is the model servo motor commonly found in radio-control cars and aeroplanes.



The servo contains a motor, gearbox, feedback potentiometer and electronic control circuit on a small printed circuit board.

The feedback potentiometer is directly connected to the output shaft. Therefore, when the motor spins, the output shaft is turned slowly by the gearbox, which in turn moves the potentiometer. Naturally movement of the output shaft is restricted to 180° as the potentiometer cannot spin continuously. The control diagram for the servo is shown below.



Note how the feedback signal from the feedback potentiometer is fed into the negative symbol of the control diagram, indicating **negative** feedback.

Past exam questions

An electronic system controls a spotlight in an art gallery. As the natural light level varies, the spotlight brightens or dims as required, in order to keep the overall light level constant. The light level at which the spotlight is triggered can be adjusted.

(*a*) Draw a control diagram for the system.



2: An electronic control system keeps the wind turbine shown below facing into the wind. The wind direction is sensed by a wind vane connected to a potentiometer. The signal from the potentiometer is compared to a second potentiometer that senses the position of the turbine head, and the head is then rotated by a motor until it is facing the wind.

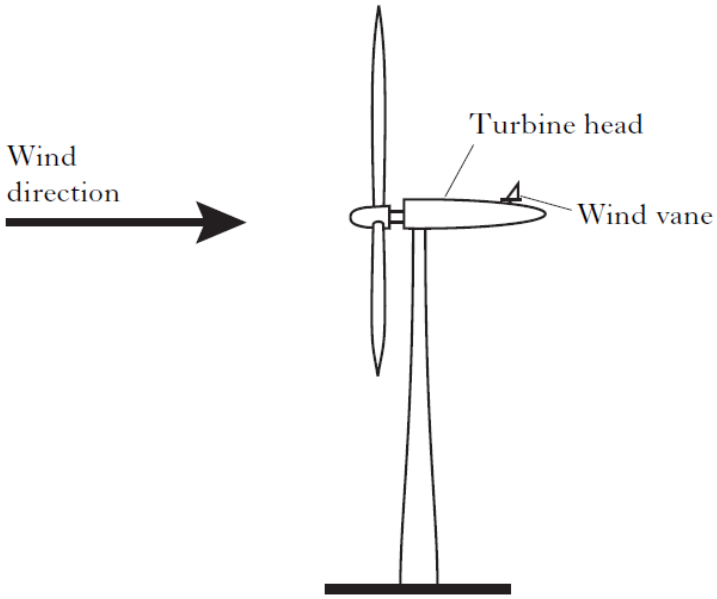


Figure Q7

(*a*) State the full name of the type of control used.

(*b*) Draw a control diagram for the direction-control system of the wind turbine.

(*c*) State which configuration of operational amplifier would be used in this electronic control system.

(*d*) (i) Draw a circuit diagram for the electronic control system, including the motor driver. Show all components and connections.

(ii) Describe how the electronic system controls the movement of the turbine head when the wind direction changes.

The hot tub shown below uses a two-state closed loop control system to regulate the water to a desired temperature.



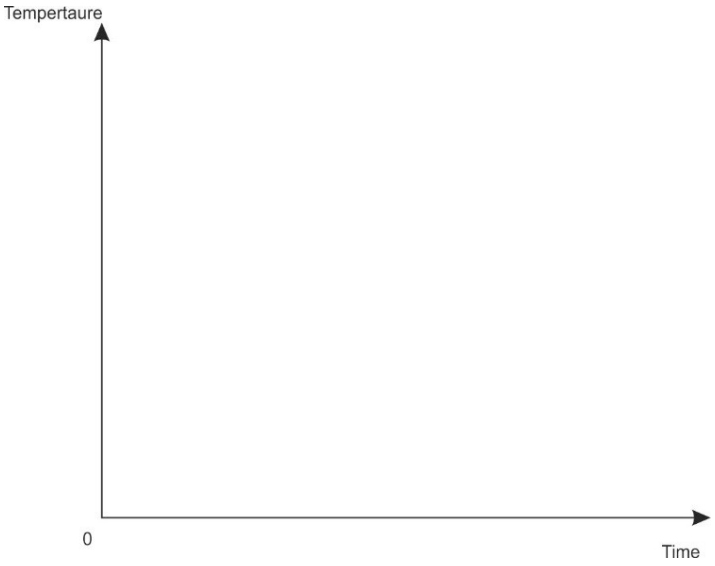
a) For the hot-tub temperature control system:

i) draw a control system diagram

ii) state the name of the op-amp configuration used in this type of control.

_____

b) Sketch a graph of the temperature against time. Show a desired temperature and show how the temperature of the water changes as it is heated from a lower temperature.



c) The control system is replaced by a closed-loop proportional control system.

i) State the name of the op-amp configuration used in this type of control.

_____

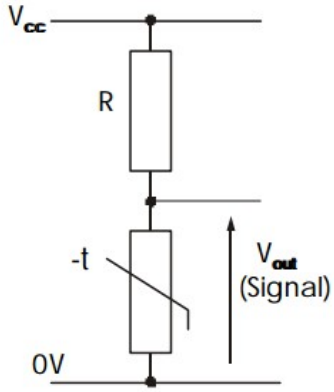
ii) Sketch, on the graph above, the ideal response of the proportional control system, as the water is heated from a temperature below the required value.

Transistor theory (As discussed in analogue electronics unit.)

Any system can be broken down into input, process and output. **Input** transducers in electronics convert one energy form into another energy form. A list of common electronic input transducers is given below.

Physical condition to be monitored	Input Transducer	Electrical property that changes
Temperature	Thermistor	Voltage

Changes in the resistance of an input transducer must be converted to changes in voltage before the signal can be processed. This is normally done by using a voltage divider circuit.



Voltage divider circuits work on the basic electrical principle that if two resistors are connected in series across a supply, the voltage load across each of the resistors will be proportional to the value of the resistors. To calculate the voltage across any one resistor in a voltage divider, we can use the following equation.

$$Signal \quad Voltage = \frac{R_2}{R_{total}} \times V_{CC}$$

Online teams task

A significant proportion of the marks at Higher come from descriptive answers to questions. Make sure you total understand how they work to allow you to answer in details.

Here is your tasks…

Using the systems and control course notes, Powerpoint presentation and your own online research, review the information and make your own jotter notes on the following:

1. What is meant by universal control?
2. What is meant by sub system control?
3. What is Open loop control? Advantages/disadvantages
4. What is Closed loop control? Advantages/disadvantages
5. What is a control diagram? Can you explain the operation of a circuit if drawn as a subsystem? It is all about 'signals' between stage. Can you sketch an error detector?
6. What is 'two state' control? Explain
7. What is 'proportional' control? What Op-amp does it use? Sketch the control graph and explain
8. Explain the different types of feedback? Can you label a positive and negative feedback diagram? Where would you use such feedbacks? Sketch the control graph and explain
9. What is 'positional' control?

These are just some of the notes you should be making. When you are reading over the course notes and you come across a task number, attempt it then continue on to the next control topic. You can miss out task 1.
This review task should take you at least 3-4 periods.